



Cameron Estridge
Computer Science

Mark Reiss
Senior SWE at Marathon

Problem / Motivation

Multiplayer games today often have high barriers to entry, including downloads, hardware requirements, and setup complexity. Pixel Coliseum addresses this by providing a lightweight, browser-based experience that enables players to instantly join and play together online without installation or complex setup.

Abstract

Pixel Coliseum is a real-time multiplayer arena game that blends pixel-art visuals with modern web technologies. Players join an online lobby, choose a class (sword, bow, or magic), and battle enemy waves in a grid-based arena. The game focuses on responsive movement and low-latency gameplay through client-side prediction and server-authoritative state synchronization.

The system features directional combat, projectile mechanics, and simple enemy AI. Pixel Coliseum highlights key principles of multiplayer game development, including real-time synchronization, scalable architecture, and interactive gameplay design.

Tech Stack

- Frontend: Phaser (TypeScript)
- Backend: Colyseus (Node.js, TypeScript)
- Database: PostgreSQL
- Networking: WebSockets



Fig. 1: Pixel Coliseum Logo



Fig. 2: Arena map

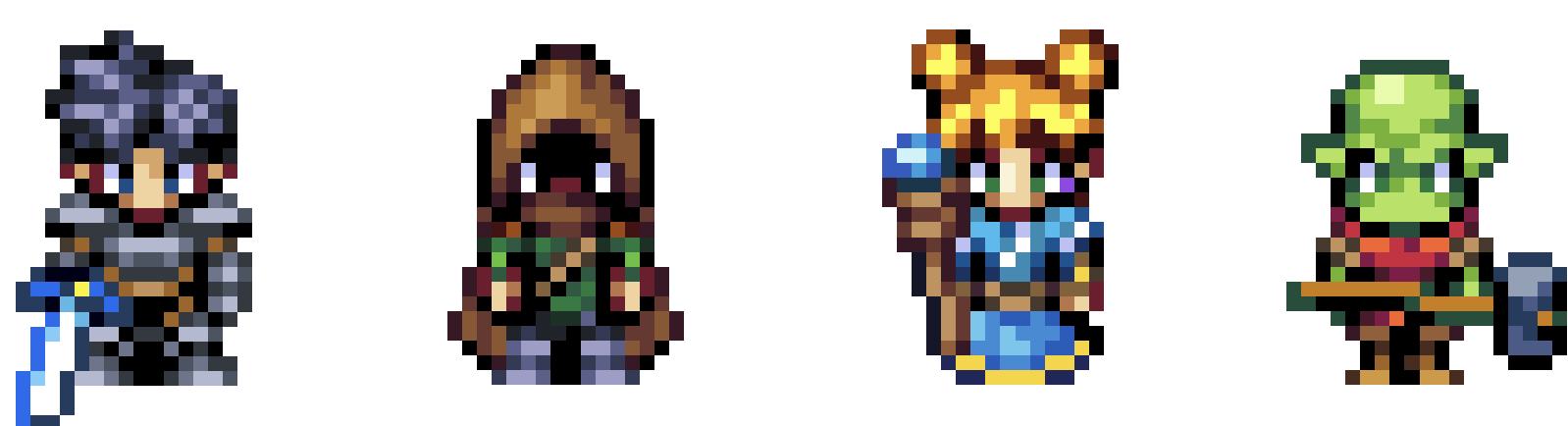


Fig. 3: Character sprites (sword class, bow class, magic class, enemy)

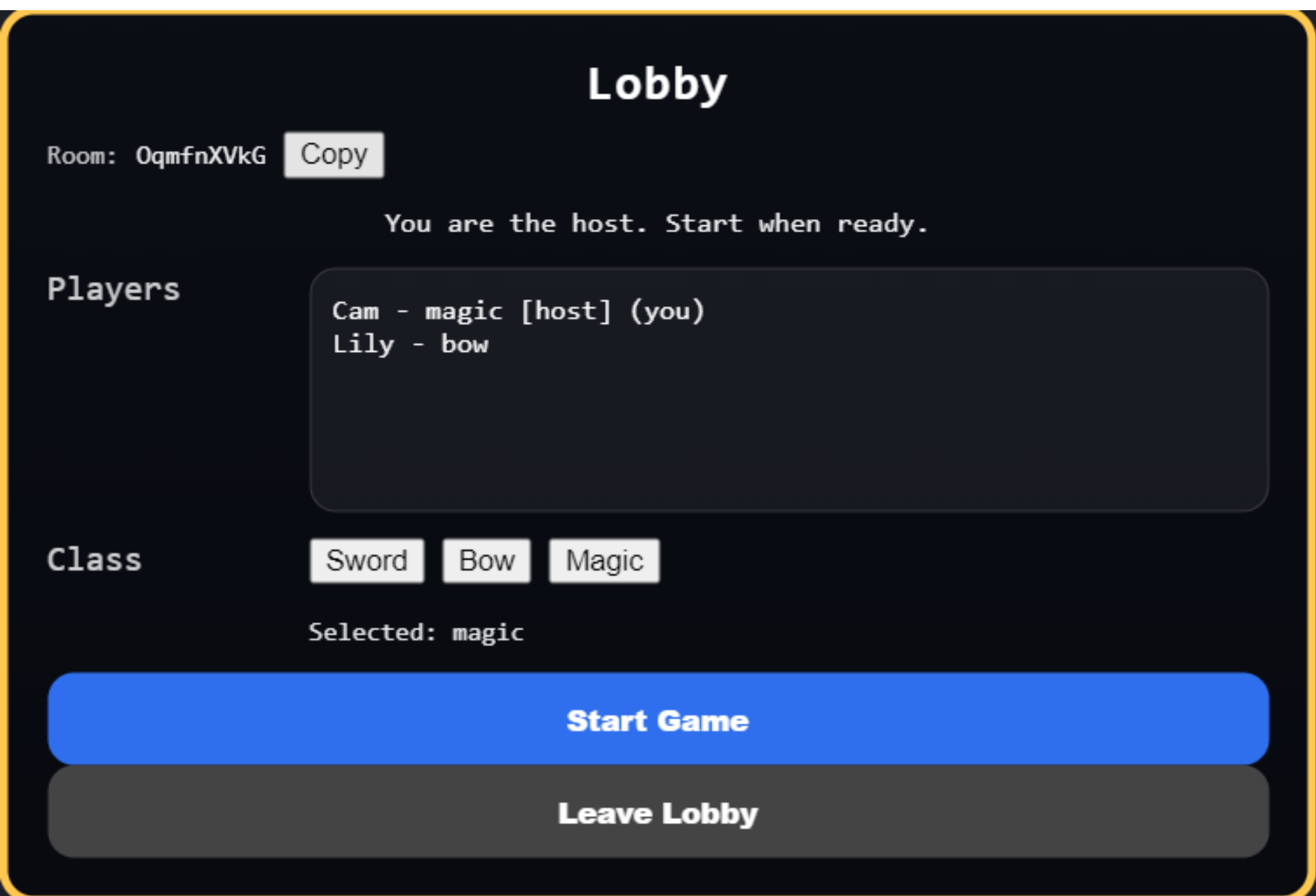


Fig. 4: Host or join other players

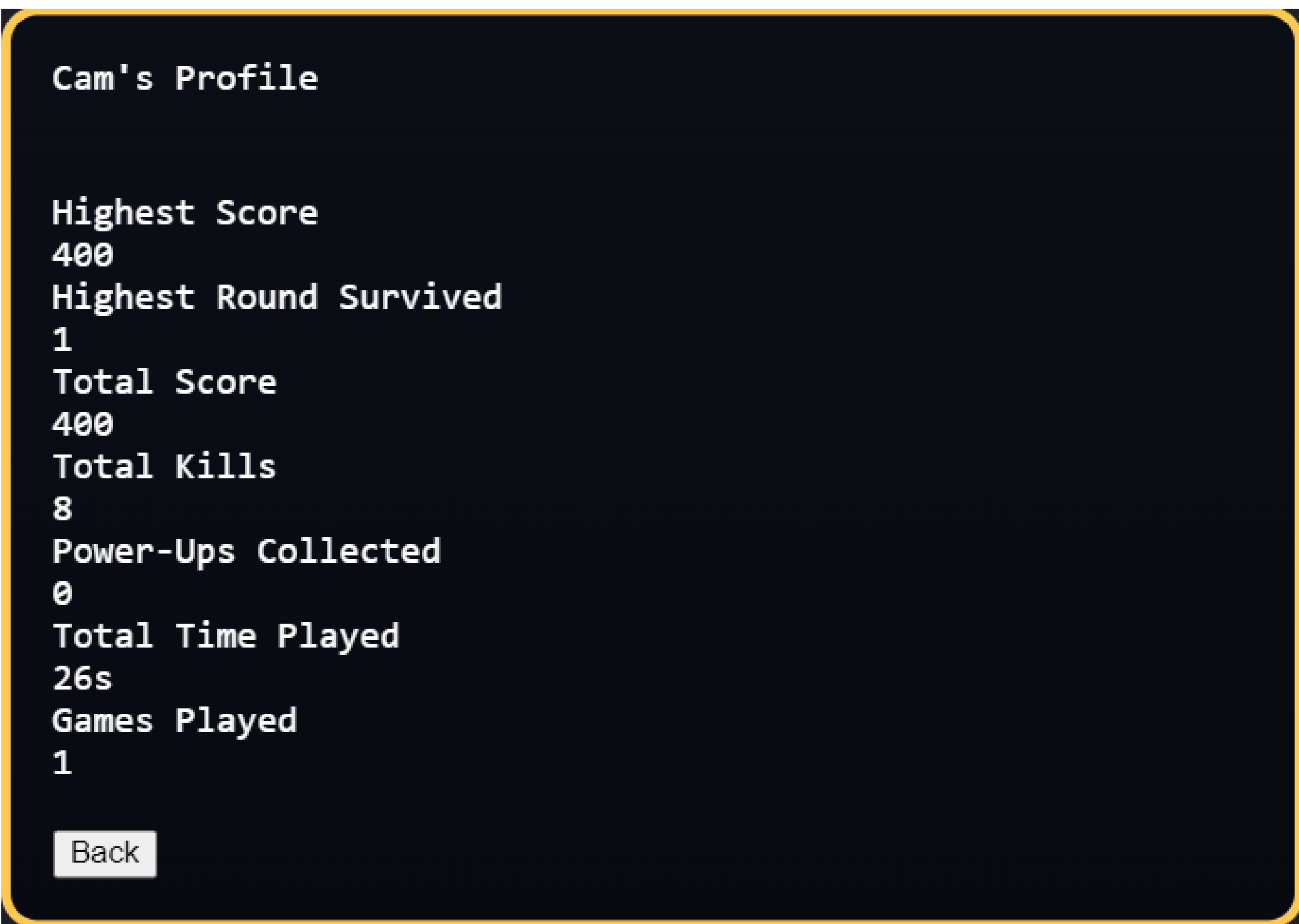


Fig. 5: Persistent player statistics

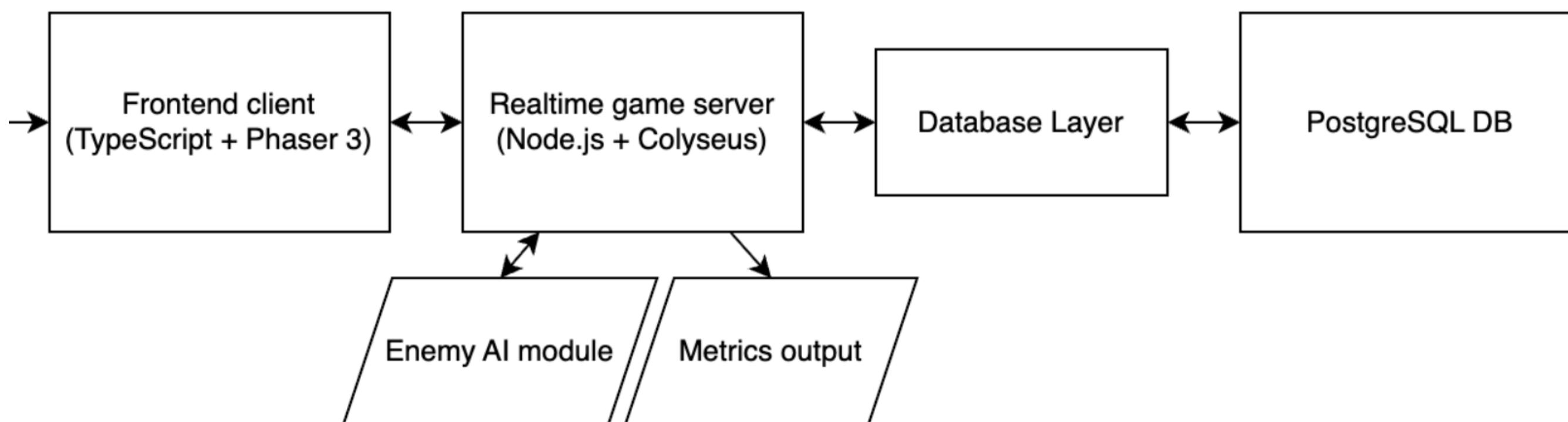


Fig. 6: Design Diagram

Technical Highlights

Multiplayer Server

- Built using Colyseus with a server-authoritative model to ensure consistent game state across clients.

AI Enemy Behavior

- Combined nearest-player targeting with A* pathfinding and a Manhattan-distance heuristic to navigate the grid around obstacles and occupied tiles.

Projectiles & Collisions

- Designed directional projectile system with server-side hit detection and travel timing.

User Authentication

- Developed secure login system with hashed passwords and session-based authentication.

Tile-Based Movement

- Enforced movement on a grid with collision checks against terrain and objects.

Animations & Combat

- Synced sprite animations with movement and attack states based on player input and facing direction.

Power-Ups System

- Implemented timed buffs (damage, speed, healing) with server-managed spawning and expiration.

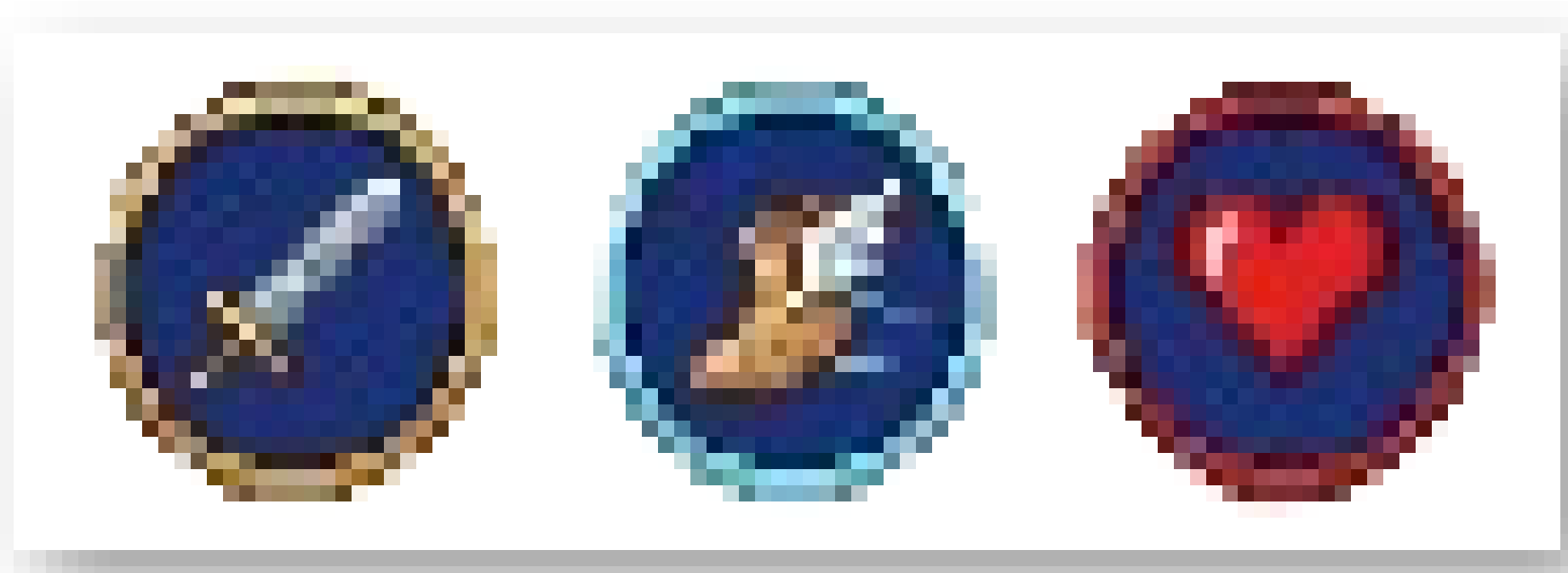


Fig. 7:Power-ups (damage, speed, heal)